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A Model for Estimating the Economic Effects of Changes in Permitting Requirements

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Abstract

This paper describes a structural model for estimating how changes in permitting requirements affect private investment. The model's framework is a representative sector composed of projects that are identical at the outset and must undergo an uncertain permitting process before production can begin. Permitting delays and outcomes are governed by idiosyncratic shocks that generate uncertainty in both the probability of a project's approval and the timing of a project's completion. The model's framework shows how longer or more uncertain permitting processes increase the user cost of capital and reduce investment, with the elasticity of investment being governed by the demand elasticity for output. The model is calibrated using sectoral data and solved numerically to evaluate the macroeconomic effects of policy changes that would alter permitting duration, costs, or approval probabilities.

Keywords: investment, regulations, time to build, user cost of capital

JEL Classification: D24, E22, Q58, R38

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1. Overview

This working paper describes a model the Congressional Budget Office developed to estimate how changes in permitting requirements would affect private investment in capital-intensive projects.¹ CBO applied this model as part of its assessment of the expected response of private investment to a permitting provision included in the House-passed version of the 2025 reconciliation act. That provision would have allowed project sponsors to pay an “opt-in” fee in exchange for faster processing times for some environmental assessments, which would not have been subject to judicial review.² Additional details about that provision and its estimated budgetary effects are available in a cost estimate.³

The model focuses on the investment behavior of capital-intensive projects within a representative sector. The sector is composed of projects that are identical at the outset (ex ante); they produce a homogeneous commodity (or good) using only capital as an input, and differences between projects arise from uncertainty during the permitting process.

Each project has a sponsor, and before a project can start earning revenue, its sponsor must obtain a permit, the process for which lasts an uncertain number of waiting periods and has an uncertain outcome. Each outcome is uncertain because a project can be approved and completed or it can be terminated. The time a project spends in the permitting queue, and the outcome after waiting in the queue, are determined by a set of idiosyncratic events, or shocks, that arrive unexpectedly and are exogenous (that is, they originate from external factors). Although project sponsors know the distribution of shocks and form rational expectations about outcomes, the time spent waiting and the final outcome of any given project are unknown at the outset.

When a project enters the permitting queue, project sponsors borrow money to pay upfront costs that represent the costs of planning and designing the project, hiring staff, securing initial financing, and making the permitting submissions, which are all unrecoverable (sunk costs). After covering those initial costs, sponsors pay a monthly fee for every period their project waits in the queue. Finally, projects incur compounding interest costs on the debt that accumulates during the waiting periods. Project sponsors have rational expectations about the market before entering and thus can weigh the costs of enduring the waiting periods against the potential revenues from producing and selling the good.

¹ Congressional Budget Office (2026), Permitting Model, www.github.com/US-CBO/permitting-model.

² Congressional Budget Office, dynamic cost estimate for H.R. 1, One Big Beautiful Bill Act (June 17, 2025; revised June 18), www.cbo.gov/publication/61486.

³ Congressional Budget Office, cost estimate for Reconciliation Recommendations of the House Committee on Natural Resources (May 19, 2025), www.cbo.gov/publication/61415. For CBO’s projected budgetary effects of the House-passed version of this bill, see estimated budgetary effects of H.R. 1, the One Big Beautiful Bill Act (June 4, 2025), <https://www.cbo.gov/publication/61461>.

The model shares features with other models in the research literature about deregulation and time to build by representing regulations as a sunk cost (Cacciatore and Fiori 2016), by modeling the option value of investment (Kalouptsi 2014 and Glancy and others 2024), and by modeling the review process as a survival process (Benyon and others 2024).

The model’s results are interpreted as changes in investment, shown as a percentage, for the portion of activity directly affected by a policy. In practice, policies may apply only to a portion of total investment within a sector. Therefore, translating the results of the model into aggregate investment effects requires scaling based on the share of activity subject to the policy. Further details on that scaling procedure, and on CBO’s general methodology for estimating responses of private investment, are available in related CBO documentation.⁴

2. A Model of the Permitting Process

Time is modeled in discrete monthly periods indexed by $t \geq 0$, and projects are indexed by i . Before a project enters the permitting queue, the project commits to the size of its capital investment, k_{it} . Also, upon entering the queue, the project incurs a one-time cost, U (for planning-related expenses) and the sponsor pays ω for every period it waits in the queue. Capital can be purchased at price p_k only after the project exits the queue successfully. A share ϵ of its obligations is financed by equity, and a share $1 - \epsilon$ is financed with debt. While waiting in the queue, the project realizes three possible random outcomes at the start of each period:

- **Exit successfully.** If a project successfully completes the permitting and building process, it earns a net present value of future profits, V_{it}^π , minus the cost of the productive capital investment, $p_k k_{it}$, minus all of its accumulated obligations, D_{it} .
- **Exit unsuccessfully.** If a project is abandoned or forced to stop the permitting process, the project earns no profits, but the sponsor is still compelled to pay the accumulated debt, $D_{it}(1 - \epsilon)$. It is not compelled to pay back equity obligations, $D_{it}\epsilon$.
- **Stay in the queue.** The present value of staying in the queue as of period t is V_{it} .

Writing this in terms of the model’s variables,

$$\begin{aligned} \text{Exit successfully:} & \quad V_{it}^\pi - p_k k_{it} - D_{it} \\ \text{Exit unsuccessfully:} & \quad -D_{it}(1 - \epsilon) \\ \text{Stay in the queue:} & \quad V_{i,t+1} \end{aligned}$$

⁴ Congressional Budget Office, “Estimating the Effects of changes in Permitting Requirements on Private Investment” (February 2026), www.cbo.gov/publication/62100.

Two random variables, θ_{it} and μ_{it} , govern which of these outcomes occurs for a project in the current period. They are idiosyncratic to the project and can take on values of 0 or 1.

Exit shock (θ_{it}): This random variable is 1 if the project exits the queue, otherwise it is 0.

Approve shock (μ_{it}): This random variable is 1 if the project is approved, otherwise it is 0.

Combining the three possible states with those shocks gives the total value of starting the project in time t :

$$V_{it} = E \left\{ \underbrace{\theta_{it}\mu_{it}[V_{it}^{\pi} - p_k k_{it} - D_{it}] - \theta_{it}(1 - \mu_{it})D_{it}(1 - \epsilon)}_{\text{Exit from queue}} + \underbrace{(1 - \theta_{it})\beta V_{i,t+1}}_{\text{Stay in queue}} \right\}. \quad (1)$$

Each project's owner has a constant discount rate of β that is between 0 and 1. Iterating equation (1) forward and applying the expectations operator yields the following:

$$V_{it} = \underbrace{\left\{ \sum_{z=0}^{\infty} \beta^z V_{i,t+z}^{\pi} g(z) \right\}}_{\substack{\text{Present value of producing} \\ = V^{\text{Total revenue}}}} - \underbrace{\left\{ \sum_{z=0}^{\infty} \beta^z k_{it} p_k g(z) \right\}}_{\substack{\text{Present value of investment} \\ = V^{\text{Capital cost}}}} - \underbrace{\left\{ \sum_{z=0}^{\infty} \beta^z D_{i,t+z} [g(z)\epsilon + f(z)(1 - \epsilon)] \right\}}_{\substack{\text{Present value of debt and equity} \\ = V^{\text{Upfront costs}}}}. \quad (2)$$

The function $f(z)$ describes the probability of any type of exit after exactly z periods in the queue:

$$f(z) = E[\theta_{i,t+z} \prod_{s=0}^z (1 - \theta_{i,t+s})].$$

The function $g(z)$ represents the probability of only a successful exit after exactly z periods in the queue, so $g(z) = E[\mu_{i,t+z} \theta_{i,t+z} \prod_{s=0}^z (1 - \theta_{i,t+s})]$. The next section describes the parameters for $f(z)$ and $g(z)$, and the following sections discuss the three components on the right side of equation (2)—total revenue, capital cost, and upfront costs—rewritten here:

$$V_{it} = V_{it}^{\text{Total revenue}} - V_{it}^{\text{Capital cost}} - V_{it}^{\text{Upfront costs}}.$$

2.1 Distribution of a Project's Time to Build

The function $f(z)$ represents the probability of an exit from the permitting queue, successful or unsuccessful, after z periods, and $g(z)$ is the probability of only a successful exit. To model the time that projects spend in the queue before an exit (survival time), this framework uses a

generalized version of the geometric distribution, called the discrete phase-type distribution (DPH), in which the survival time is a Markov chain sequence of stages.

More stages in the Markov chain of $f(z)$ allow the distribution to match more moments in the data. For example, with more than one stage, the model can match both the mean and median of an observed distribution, not just the mean. Also, the multiple stages represent distinct permitting requirements, or phases of the project's life cycle, such as planning, permitting, and construction. For cases that have only one stage, the distribution for the time to build is a simple geometric distribution, and the Markov chain is reduced to a set of independent Bernoulli trials.

The distribution for $f(z)$ is defined below for $z \geq 0$. Letting m be the number of stages, A is an $m \times m$ transition matrix that represents the probabilities of transitioning among one of m stages of the permitting process:

$$f(z) = a_0' A^z (S + H).$$

The matrices S and H are $m \times 1$ matrices that represent the probability of exiting successfully (S) or unsuccessfully (H) given that the project started that period in one of the m -states. Finally, a_0 is an $m \times 1$ matrix that describes how projects are initially distributed across the m -states. The probability of a successful exit, $g(z)$, is the probability of exiting and not having an unsuccessful exit, so $g(z) = a_0' A^z S$.

In this model, the number of stages is set to two ($m = 2$), so the matrices are the following:

$$a_0 = \begin{bmatrix} \alpha_1 \\ 1 - \alpha_1 \end{bmatrix}, A = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix}, S = \begin{bmatrix} (1 - \alpha_{11} - \alpha_{12})s_1 \\ (1 - \alpha_{22} - \alpha_{21})s_2 \end{bmatrix}, H = \begin{bmatrix} (1 - \alpha_{11} - \alpha_{12})(1 - s_1) \\ (1 - \alpha_{22} - \alpha_{21})(1 - s_2) \end{bmatrix}.$$

The parameters here can be interpreted as follows: When i or j is equal to 1 or 2, α_{ij} is the probability of transitioning to stage j given the project started that period in stage i , s_i is the probability of a successful exit from stage i in that period, and α_i is the probability that the project started the permitting process in stage i . Acceptable values for those parameters are restricted to between 0 and 1. Refining the values of the parameters is discussed in Section 4: Calibration.

2.2 Present Value of Upfront Costs

Obligations accumulate while the project waits in the queue. During the initial period of entry, the project sponsor purchases an initial investment that is sunk, U . In later periods, the project incurs additional costs, ω , and rolls over obligations from previous periods. For each period, the project sponsor pays interest on debt and issues dividends on equity shares at rates r_d and r_e , respectively. This paper defines $R \equiv 1 + r_e \epsilon + r_d(1 - \epsilon)$, and r_e and r_d are set larger than $1/\beta - 1$. The parameter ϵ represents the share of financing that comes from equity:

$$\begin{aligned}
B_{i0} &= U \\
B_{i1} &= B_{i0}R + \omega \\
&\dots \\
B_{iz} &= B_{i,z-1}R + \omega.
\end{aligned}$$

Plugging these into each other and iterating yields

$$B_{iz} = R^z \left(U + \frac{\omega}{R-1} \right) - \frac{\omega}{R-1}.$$

The project exits after z periods and its sponsor starts making regular payments to lenders and shareholders. The project regularly pays off a share ψ of its obligation for the current period. However, while doing so, the project incurs additional interest and dividend costs on the remaining obligations. Therefore,

$$B_{i,z+1} = B_{iz}R - \psi B_{iz}R.$$

The parameter ψ is defined conveniently so that projects always have the same debt-to-capital ratio:

$$\frac{B_{i,z+1}}{B_{iz}} = R(1 - \psi) = 1 - \delta.$$

So ψ is

$$\psi = \frac{R+\delta-1}{R}.$$

After n periods of producing a good, the project's remaining obligation in period n is

$$B_{i,z+n} = B_{iz}(1 - \delta)^n.$$

The present value of all future payments is the discounted sum of all future payments:

$$\begin{aligned}
D_{iz} &= \sum_{n=0}^{\infty} \beta^n \psi B_{i,z+n} R \\
&= B_{iz} p_d,
\end{aligned}$$

where p_d is the added cost of carrying obligations while producing:

$$p_d = \frac{\beta(1-\delta)(R-1+\delta)}{1-\beta(1-\delta)}. \quad (3)$$

Those payments start when the project exits the queue. The project sponsor is only obligated to pay equity to investors if the project exits successfully—and the project is obligated to pay the debt-financed portion for any type of exit. Therefore, the present value of these payments is

$$V_{it}^{\text{Upfront costs}} = \sum_{z=0}^{\infty} \beta^z \left[\underbrace{D_{i,t+z} \epsilon g(z)}_{\text{Equity payout}} + \underbrace{D_{i,t+z} (1 - \epsilon) f(z)}_{\text{Debt payout}} \right].$$

Simplifying, this is

$$V_{it}^{\text{Upfront costs}} = U p_u + \omega p_\omega,$$

where coefficients p_u and p_ω are equal to

$$p_u = p_d a'_0 [I - \beta A R]^{-1} [S + (1 - \epsilon) H] \quad (4)$$

$$p_\omega = p_d a'_0 ([I - \beta A R]^{-1} - [I - \beta A]^{-1}) (R - 1)^{-1} [S + (1 - \epsilon) H]. \quad (5)$$

Those coefficients represent the added cost of financing the sunk (U) and per-period costs (ω).

2.3 Present Value of Capital Cost

Successful projects will buy capital at the aggregate-good price, normalized to 1. Tax and depreciation deductions are applied to the capital purchase. τ is the marginal tax rate on income, Z represents the value of depreciation deductions, and ξ is used in calibration and represents other deductions or taxes that may apply. The price of capital after these deductions is $p_k = 1 - \tau Z \xi$.

Similar to the upfront costs, the capital is financed by debt and equity. Using the definition of p_d given by equation (3), the future value of payments on productive capital k_{it} is given by $p_d p_k k_{it}$. The term $V_{it}^{\text{Capital cost}}$ in equation (2) is

$$\begin{aligned} V_{it}^{\text{Capital cost}} &= \sum_{z=0}^{\infty} \beta^z p_d p_k k_{it} g(z) \\ &= k_{it} \bar{p}_k, \end{aligned} \quad (6)$$

where

$$\bar{p}_k = p_d p_k a'_0 (I - \beta A)^{-1} S.$$

2.4 Present Value of Total Revenue

Successful projects produce a good with constant returns to scale production. They buy k_{it} upon a successful exit, and capital starts depreciating at rate δ . Then $j \geq 1$ periods after exit, the project produces y_{ijt} using the remaining capital $(1 - \delta)^j k_{it}$:

$$y_{ijt}(h) = (1 - \delta)^j k_{it}.$$

The project sells output at price P_t in time t , and there are no marginal costs:

$$\pi_t(y_{ijt}) = y_{ijt}P_t.$$

Given a successful exit from the queue at time t , the present value of future profits is

$$\begin{aligned} V_{it}^\pi &= \beta\{\pi_{t+1}(y_{ijt}) + V_{i,t+1}^\pi\} \\ &= k_{it} \sum_{z=1}^{\infty} P_{t+z} \beta^z (1 - \delta)^{z-1}. \end{aligned}$$

Written recursively, the term x_t is defined to represent the present value of capital after a successful exit,

$$x_t = \beta P_{t+1} + \beta x_{t+1}(1 - \delta), \quad (7)$$

so that V_{it}^π can be written as $V_{it}^\pi = k_{it}x_t$. Therefore $V_{it}^{\text{Total revenue}}$ in equation (2) can be written as

$$\begin{aligned} V_{it}^{\text{TotalRevenue}} &= \sum_{z=0}^{\infty} \beta^z V_{i,t+z}^\pi g(z) \\ &= k_{it} \sum_{z=0}^{\infty} \beta^z x_{t+z} g(z) \\ &= k_{it} a'_0 \chi_t, \end{aligned}$$

where χ_t is

$$\chi_t = Sx_t + \beta A\chi_{t+1}. \quad (8)$$

2.5 Project Entry and Investment

Projects enter the permitting queue until the value of entering is 0:

$$V_{it} \geq 0, \quad (9)$$

and using the results from the sections above to write V_{it} ,

$$k_{it} a'_0 \chi_t - k_{it} \bar{p}_k - Up_u - \omega p_\omega \geq 0. \quad (10)$$

In equilibrium, projects choose capital so that equation (10) holds with equality. After rearranging,

$$k_{it} = \frac{Up_u + \omega p_\omega}{a'_0 \chi_t - \bar{p}_k}. \quad (11)$$

The variable k_{it} is the level of investment for each period of the model that ensures that the free-entry condition in equation (9) holds.

2.6 Aggregation

Because project sponsors view market prices as outside their control (they are price-takers), face sunk entry costs, and operate with constant returns to scale production, the model does not determine the number of entrants endogenously and does not have a finite, nondegenerate

distribution of project sizes.⁵ Instead, the mass of entrants for each period is set at a constant value, n :

$$k_t = nk_{it} = \int_{i \in \Omega_t} k_{it} di,$$

and Ω_t is the set of i -entrants that enter in time t . The number of projects in stage v of m possible stages is

$$N_t^v = \sum_{q=1}^m N_{t-1}^q \alpha_{qv} + a_{0v}n,$$

with a_{0v} being the v th row of a_0 , and α_{qv} being the entry in row q and column v of A . The average amount of investment for each project in stage v , $\bar{k}_t^v$, is the weighted average of investment for all entering projects:

$$\bar{k}_t^v = \sum_{q=1}^m \phi_{qv} \bar{k}_{t-1}^q + a_{0v}k_{it},$$

where each weight is

$$\phi_{qv} = \frac{N_{t-1}^q \alpha_{qv}}{N_t^v}.$$

Rewriting the equation and stacking across m -states yields

$$\Delta_t' = \Delta_{t-1}' A + a_0' k_{it}. \quad (12)$$

where Δ_t is an $m \times 1$ vector with values in each row, v , equal to $\Delta_t^v = N_t^v \bar{k}_t^v$. Projects that successfully exit purchase capital and invest, so the total investment for each period is

$$I_t = \Delta_t' \times S. \quad (13)$$

Depreciation starts once the project starts producing, so the total output for each period is

$$Q_t = Q_{t-1}(1 - \delta) + I_t. \quad (14)$$

The demand curve for Q_t is represented by a function for constant elasticity of substitution that satisfies the condition that the total supplied quantity of the good equals the demanded quantity:

$$P_t = \gamma(Q_{t-1})^{-1/\sigma}, \quad (15)$$

where $\sigma > 0$ and $\gamma > 0$ are parameters.

⁵ See Dixit and Stiglitz (1977) for the relevant discussion of firm entry, sunk costs, and price taking.

3. The User Cost of Capital

In this section, CBO derives several expressions used to calibrate the elasticity of investment with respect to the user cost of capital (UCC). The section only derives conditions for the steady state of the model, and thus, t -subscripts are dropped for convenience. The free-entry condition in equation (9) implies that, for a project entering in any period, the present value of the returns equals the present value of capital and upfront costs:

$$V_i^{\text{Total revenue}} = V_i^{\text{Capital cost}} + V_i^{\text{Upfront costs}} = V_i^{\text{Total cost}}. \quad (16)$$

Scaling both sides by k_{it} , the rate of return at which the marginal project is indifferent to entering is defined as V_i^{Enter} . Also, for convenience, V_i^{Succeed} is defined as the return on a project if it is successful:

$$\frac{V_i^{\text{Total revenue}}}{k_i} = \frac{V_i^{\text{Total cost}}}{k_i} \equiv V_i^{\text{Enter}} = \sum_{s=0}^{\infty} \beta^s g(z) V_i^{\text{Succeed}}. \quad (17)$$

Because some projects fail and successful projects do not earn revenue while in the permitting queue, the value of a project before entry is lower than its value after a successful exit.

Suppose the project successfully exits and starts earning a consistent return of $\rho + \delta$ in each period. That return compensates the sponsor for the depreciation of capital, δ , and provides a market rate of return net of depreciation, ρ . The UCC is defined as $\rho + \delta$, and the present value of those returns after adjusting for taxes (τ) is

$$\begin{aligned} V_i^{\text{Succeed}} &= \beta(1 - \delta)(\rho + \delta)(1 - \tau) + \beta^2(1 - \delta)^2(\rho + \delta)(1 - \tau) + \dots \\ &= (\rho + \delta) \frac{\beta(1 - \delta)(1 - \tau)}{1 - \beta(1 - \delta)}. \end{aligned} \quad (18)$$

Using the definition for $g(z)$, V_i^{Enter} is

$$\begin{aligned} V_i^{\text{Enter}} &= \sum_{s=0}^{\infty} \beta^s g(z) V_i^{\text{Succeed}} \\ &= (\rho + \delta) \frac{\beta(1 - \delta)(1 - \tau)}{1 - \beta(1 - \delta)} a'_0 (I - \beta A)^{-1} S, \end{aligned} \quad (19)$$

where the second line of equation (19) follows because $\sum_{s=0}^{\infty} \beta^s A^s = (I - \beta A)^{-1}$. Finally, using the results that $V_i^{\text{Total cost}} = k_i \bar{p}_k + Up_u + \omega p_\omega$, and rearranging,

$$\frac{k_i \bar{p}_k + Up_u + \omega p_\omega}{k_i} = (\rho + \delta) \frac{\beta(1 - \delta)(1 - \tau)}{1 - \beta(1 - \delta)} a'_0 (I - \beta A)^{-1} S.$$

Then using equation (11), the UCC is

$$UCC \equiv \rho + \delta = a'_0 \chi \left(\frac{1 - \beta(1 - \delta)}{\beta(1 - \delta)(1 - \tau) a'_0 (I - \beta A)^{-1} S} \right). \quad (20)$$

In the steady state, it is possible to show that the elasticity of investment with respect to the UCC is equal to the price elasticity of demand for the good, which is the parameter $-\sigma$ used in the demand curve in equation (15):

$$\frac{dI}{I} \frac{UCC}{dUCC} = -\sigma.$$

In research literature about this UCC elasticity, the appropriate calibration for σ depends on the model's setting. In this model, Q_t represents capital goods in a partial equilibrium setting in which labor, wages, interest rates, and an aggregate good's quantity and price are determined outside of the model. Capital firms purchase Q_t as installed capital, and the elasticity of substitution between capital and labor (or some other input) is equal to $-\sigma$.

This interpretation of σ is consistent with what Chodorow-Reich (2025) refers to as “the generalized Hall and Jorgenson (1969) user cost elasticity”—or the elasticity of substitution between capital and labor in the production function in which capital is used. Therefore, $\sigma = 0.7$, a value consistent with this paper's interpretation and with CBO's modeling of this elasticity elsewhere.⁶

4. Calibration

Table 1 shows the key parameters of the model with descriptions and context about how CBO determined the values.

⁶ Congressional Budget Office, “Key Methods That CBO Used to Estimate the Macroeconomic Effects of the 2017 Tax Act” (supplemental material for *The Budget and Economic Outlook: 2018 to 2028*, April 2018), www.cbo.gov/publication/53651. Chodorow-Reich (2025) also provides a detailed discussion of CBO's use of this elasticity.

Table 1: Model Parameters

Parameter	Description	Source
δ	Depreciation rate, monthly	The Congressional Budget Office's CapTax Model: ⁷
β	Long-run rate of return on risk-free debt, monthly	
τ	Marginal tax rate on earnings	
ρ	Return on investment net of depreciation, monthly	
R	The weighted return on debt and equity for each Bureau of Economic Analysis (BEA) sector, monthly	
Z	The share of depreciation deductions for each BEA sector	Calibrated so ρ matches CapTax's return on investment after depreciation
ξ	Other taxes and deductions that may apply for each BEA sector	
σ	Price elasticity of demand for the project (also equivalent to the long-run elasticity of investment with respect to the user cost)	Congressional Budget Office (2018)
U	Sunk cost of entering the permitting queue	Calibrated to match the ratio of fixed assets to total assets for each BEA sector ⁸
ω	The per-period constant cost of waiting in the permitting queue	Calibrated to results from Trebbi and Zhang (2022)
α_{11}, α_{22}	Probabilities of remaining in stage 1 (stage 2) next period, given that the project began that period in stage 1 (stage 2)	Calibrated to match the mean and median of the time to build distribution for each BEA sector
α_1	Probability of starting in the queue from stage 1	Set to 1 for all BEA sectors
α_{12}	Probability of transitioning to stage 2, having started that period in stage 1	Set to $1 - \alpha_{11}$ for all BEA sectors
α_{21}	Probability of transitioning to stage 1, having started that period in stage 2	Set to a small value consistent across BEA sectors
s_1	Probability of success in any period, having started that period in stage 1 and having exited in that period	Set near 1 following results in Glancy and others (2024), which suggest that projects are likely to succeed conditional on an exit

⁷ Paul Burnham and Dorian Carloni, *CBO's Model for Estimating the Effect That Federal Taxes Have on Capital Income From New Investment*, Working Paper 2022-01 (Congressional Budget Office, February 2022), www.cbo.gov/publication/57429.

⁸ BEA sector 4820 (rail transportation) is an outlier with a high fixed asset ratio. To ensure a reasonable calibration for that sector, the fixed asset ratio from BEA sector 4860 (pipeline transportation) is used in its place.

5. Solution

In summary, the model has seven variables (x_t , χ_t , P_t , Q_t , I_t , k_t , Δ_t) and seven equations (7, 8, 11, 12, 13, 14, 15). The model is solved in Dynare 6.2 using the perfect foresight solver with default settings.⁹ When the dimensions of A subject to an unexpected policy change are greater than 1 (that is, $m > 1$), the model's dynamics depend on the initial conditions, and there is a random walk component. Because the benchmark has $m = 2$, this paper uses an approach like those discussed in Schmitt-Grohé and Uribe (2003) to make an ad hoc adjustment in the model. The adjustment has a negligible effect on the solution but grants long-run stability by eliminating the model's random walk. The adjustment defines an autoregressive process in the sunk cost U , so that sunk costs rise as expected capital demand increases relative to steady-state capital. Specifically, U is allowed to vary over time according to the process $U_{t+1} = U(1 - \theta) + U_t\theta(Q_t/\bar{Q})$, where U is the calibrated sunk cost parameter, $\bar{Q}$ is the steady-state value of capital output, and θ controls the deviation from U . To minimize the effect of this ad hoc correction to the results, θ is set to $1E - 10$.

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⁹ Documentation for Dynare is available at <https://www.dynare.org>.

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